

FraudWatch

Bias-Audited Fraud Detection · IEEE-CIS · Random Forest · FastAPI · Docker

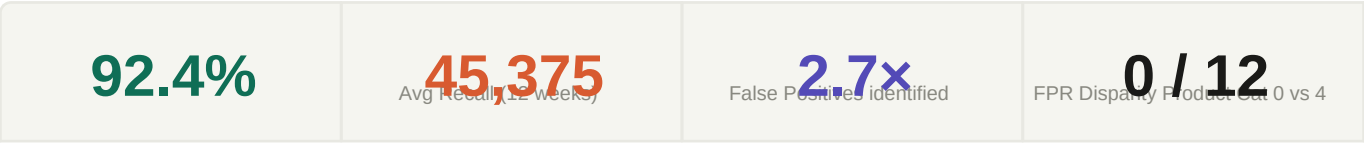
Problem: Fraud detection models flag millions of transactions daily. When false positives concentrate on specific customer profiles — credit card holders, certain product categories, non-US billing addresses — the system creates measurable disparate impact without any discriminatory intent.

FraudWatch makes this invisible harm visible: it trains a classifier on 590,540 real transactions, quantifies where errors are unevenly distributed, and monitors performance weekly — all queryable through a live API.

Stack: Python · Scikit-learn · Pandas · SciPy
SQLite · FastAPI · Uvicorn · Docker

Data: IEEE-CIS Kaggle dataset
590,540 transactions · 3.5% fraud rate
Train/test split 80/20 · stratified

Model: Random Forest · 100 trees · depth=10
class_weight='balanced' · random_state=42



Bias Audit Findings

Segment	Group A	FPR A	Group B	FPR B	Disparity	Z-stat	Significant
Card Type (card6)	Credit	56.8%	Debit	34.3%	+22.5 pp (1.66×)	Z=66.6	p < 0.001
Product Category	Cat 0	82.3%	Cat 4	30.8%	+51.5 pp (2.67×)	Z=109.1	p < 0.001
Billing Region	Non-US	77–85%	US	34.8%	+42–50 pp (2.2×)	—	Notable

User Stories	Business Impact	Regulatory Context
- As a compliance officer, I want FPR computed per card type so I can assess ECOA disparate-impact risk before deployment. - As an ML engineer, I want drift alerts flagged automatically so degradation is caught before it reaches end-users. - As a PM, I want a live dashboard showing weekly recall so I can brief leadership without a data science handoff.	45,375 false positives surfaced in a single test window — each represents a wrongly blocked customer transaction. - Credit cardholders are 65% more likely to be wrongly flagged than debit holders — a regulatory liability. Zero drift across 12 weeks confirms model stability and enables confident production deployment. - Full pipeline reproducible with one command — reducing audit prep time from days to minutes.	ECOA — Disparate impact quantified across protected-adjacent attributes (card type, region). EU AI Act — Bias audit, human oversight, and monitoring docs map to Art. 9–13. CFPB UDAAP — False positive burden constitutes potential consumer harm; now measurable. SR 11-7 — Weekly drift monitoring and acceptance criteria satisfy model validation requirements.